# Interval notation

**1** Write the interval notation for each of the following pictured inequalities:

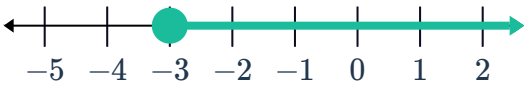
**a**



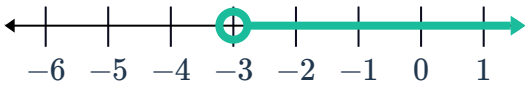

**b**



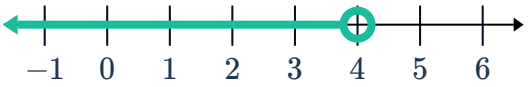

**c**




**d**




**e**




**2** For each of the following inequalities express the solutions of the inequality using interval notation:

**a** $x < -7$ **b** $x \leq 6$ **c** $x \geq -4$ **d** $-7 \leq x \leq 3$

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## Domain and range

**3** For each of the following relations:

**i** Find the domain.

**ii** Find the range.

**iii** Determine whether the relation is a function or not.

**a** $\{(8,5),(1,3),(3,4),(9,1),(2,9)\}$

**b** $\{(2,3),(3,8),(6,1),(8,7),(3,2)\}$

**c** $\{(4,4),(8,9),(2,8),(7,9),(3,1)\}$

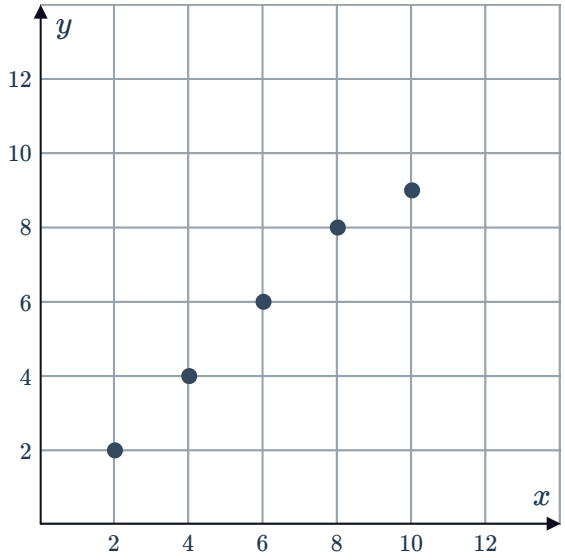
**d**

| $x$ | 1 | 6 | 3 | 8 | 2 |
|---|---|---|---|---|---|
| $y$ | 3 | 2 | 7 | 1 | 2 |

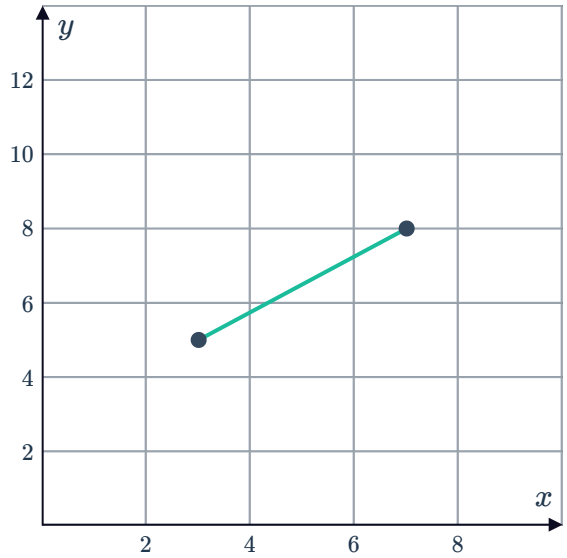
**e**

| $x$ | 7 | 7 | 8 | 5 | 3 |
|---|---|---|---|---|---|
| $y$ | 1 | 9 | 3 | 2 | 6 |

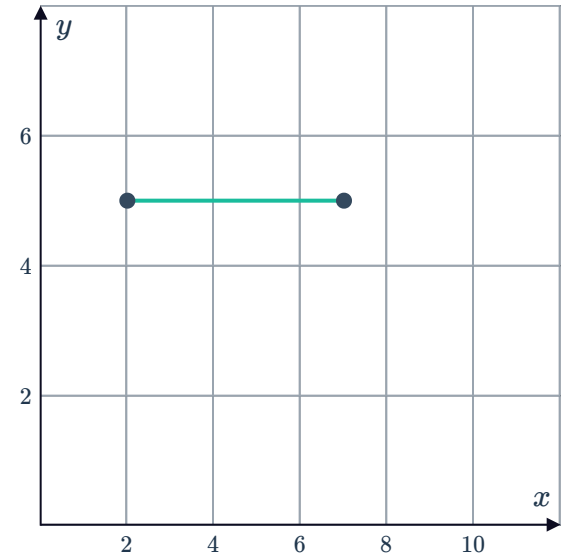
**f**




**g**

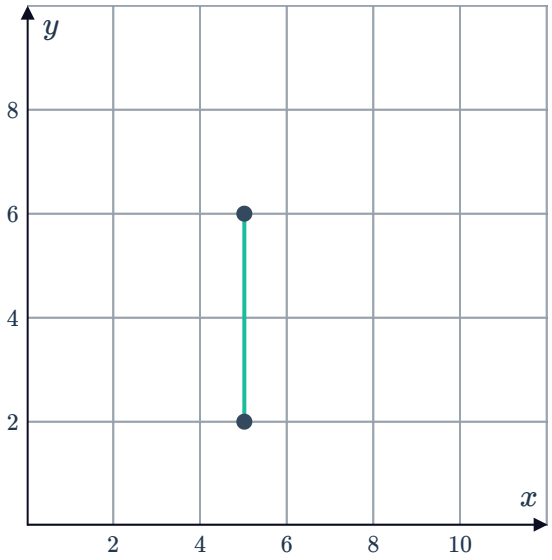



**h**

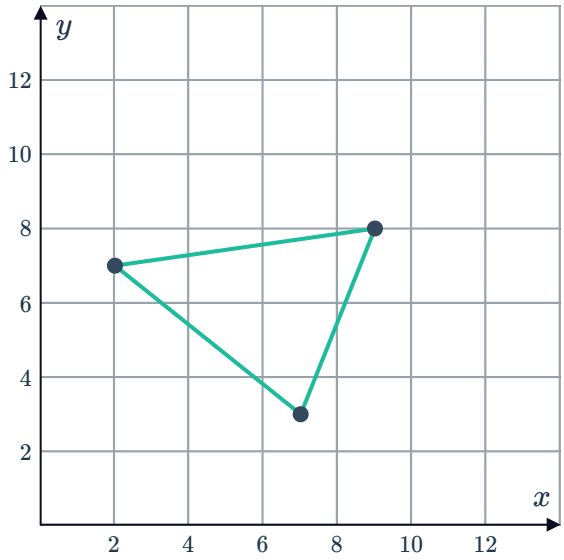



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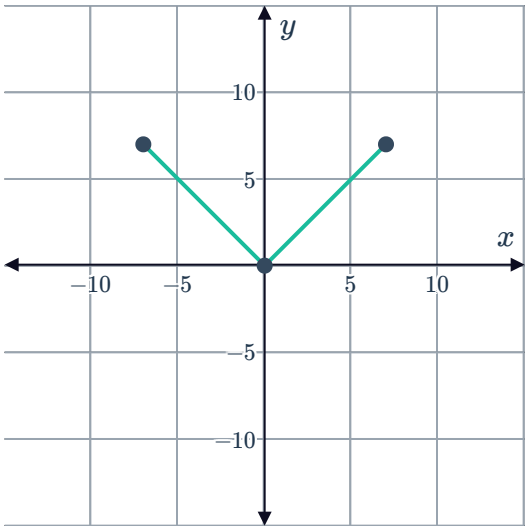
i



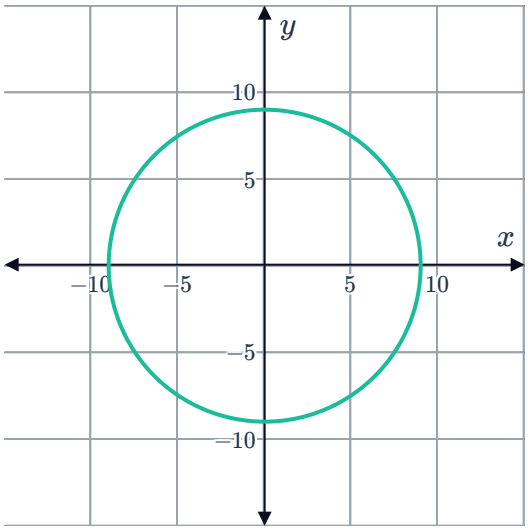

j



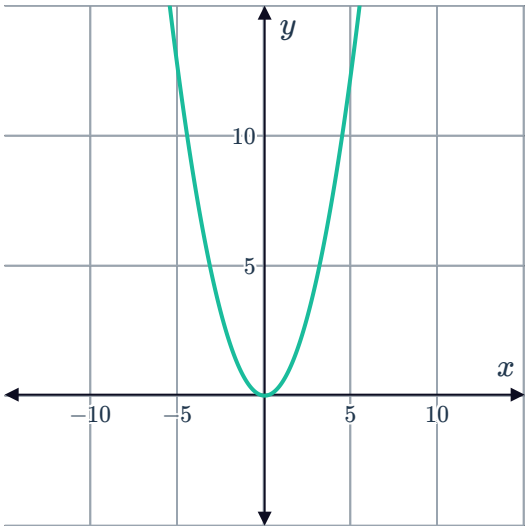

k



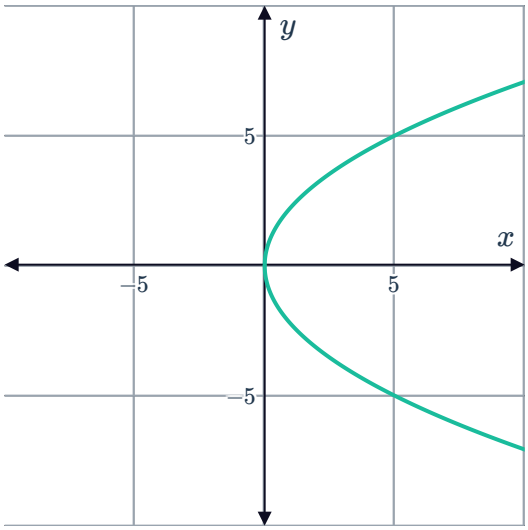

l




m



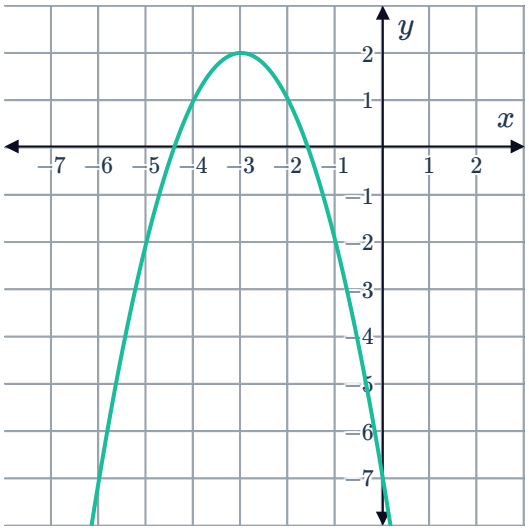

n



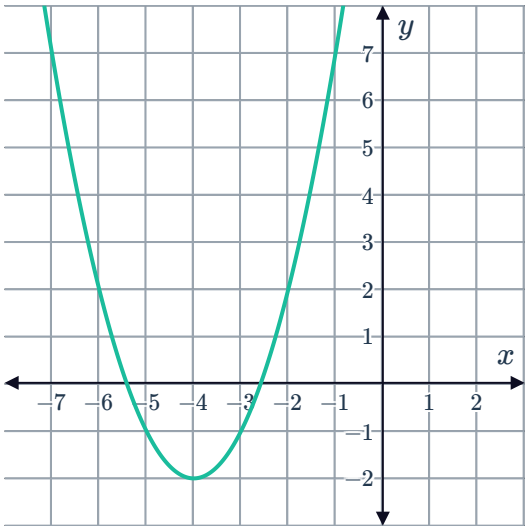

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**4** State the domain and range of the following functions:

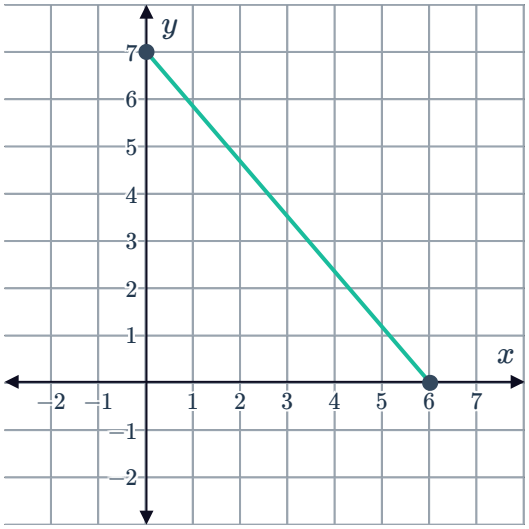
**a**




**b**




**c**



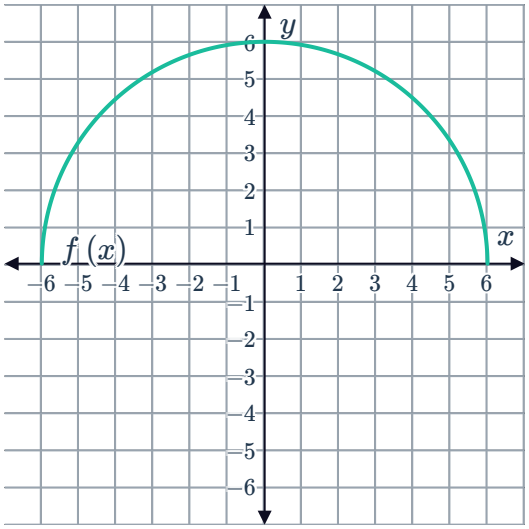

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**5** State the domain of the following functions:

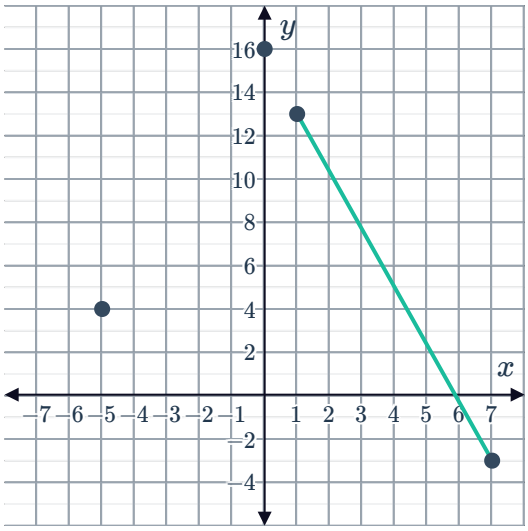
**a** $f(x) = -3x - 1$

**b** $f(x) = 8x + 8$

**c**




**6** Consider the function $f$ graphed. Use the graph to determine whether each of the given statements are true or false:




**a** The domain is $[-5, 7]$

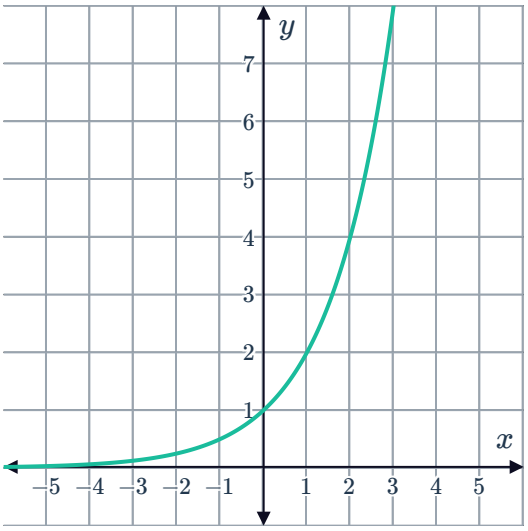
**b** The range is $[-3, 16]$

**c** $f(1) - f(7) = 16$

**d** $f(0) = 16$

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**7** Consider the graph of $y = 2^x$:




**a** State the $y$-intercept of the graph.

**b** Does the graph have an $x$-intercept?

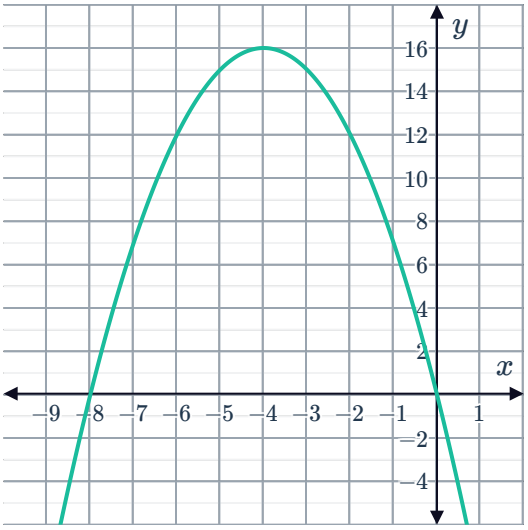
**c** State the domain of the function.

**d** State the range of the function.

**e** Find the value of $y$ when $x = 7$.

**f** Find the value of $x$ when $y = 256$.

**8** Consider the graph of the function shown:

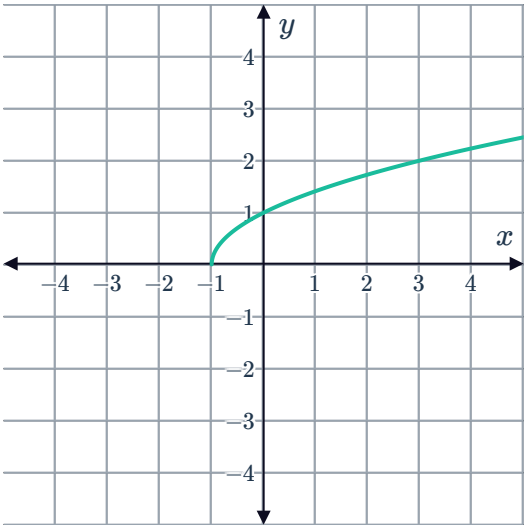



**a** State the maximum value of the graph.

**b** State the range of the function.

**c** State the domain of the function.

**9** Consider the graph of the function $f(x) = \sqrt{x+1}$:




**a** State the domain of the function.

**b** Is there a value in the domain that can produce a function value of $-2$?

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**10** Consider the function $f(x) = (x-2)^2$, for all real $x$.

**a** Graph the function.

**b** Is there a value in the domain that can produce a function value of $-9$?

## Continuity

**11** A function $f(x)$ is known to be continuous on the interval $(-\infty, \infty)$.

**a** Is $f(x)$ continuous over all smaller intervals?

**b** Suppose the function is $f(x) = 3x + 8$. State whether $f(x)$ is continuous on the following intervals:

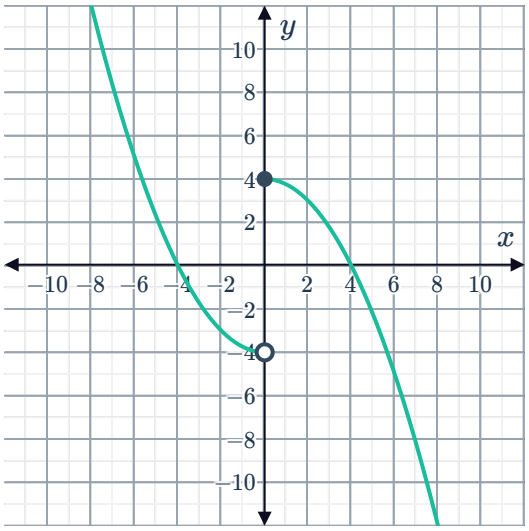
**i** $[4, 6]$ **ii** $(-7, -1)$ **iii** $(0, \infty)$

**12** Consider the function $f(x) = 4x^3 - 9x^2 - 2x + 1$. State whether $f(x)$ is continuous over the following intervals:

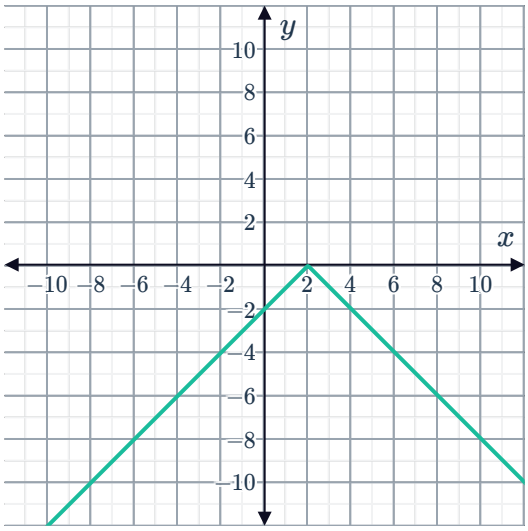
**a** $(-\infty, \infty)$ **b** $[9, 15]$

**13** State whether the following functions are continuous:

**a**

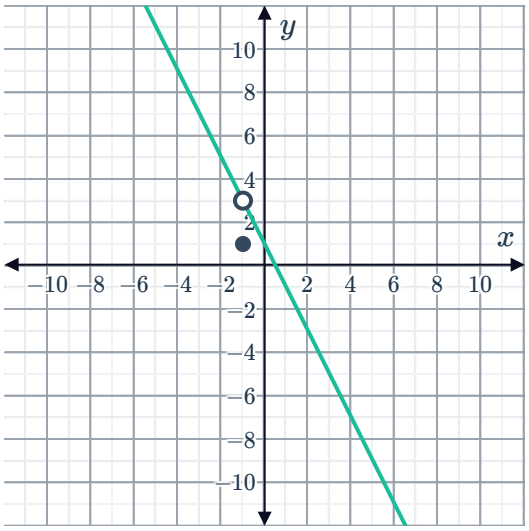



**b**

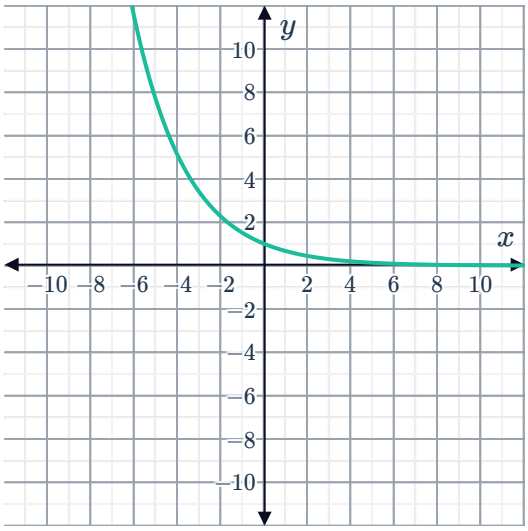



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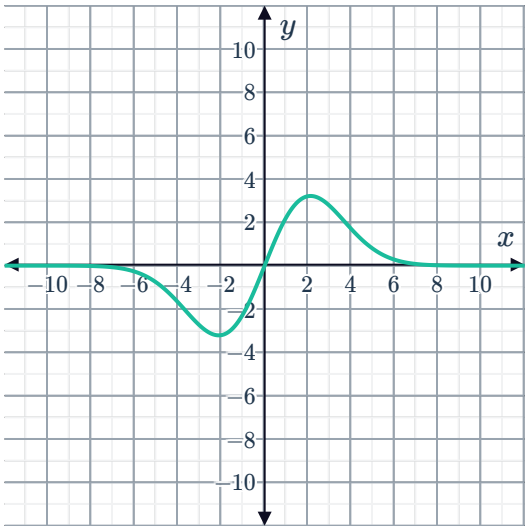
c



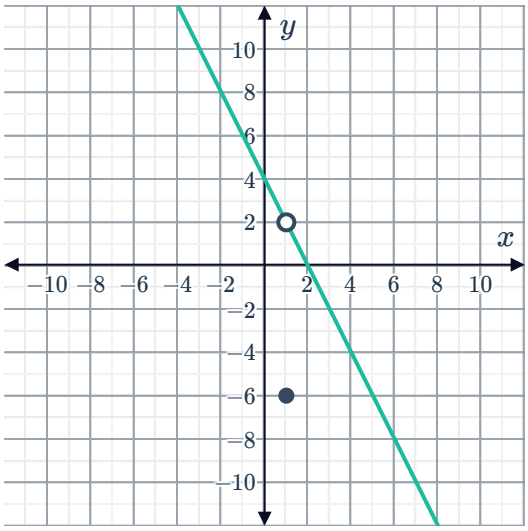

d



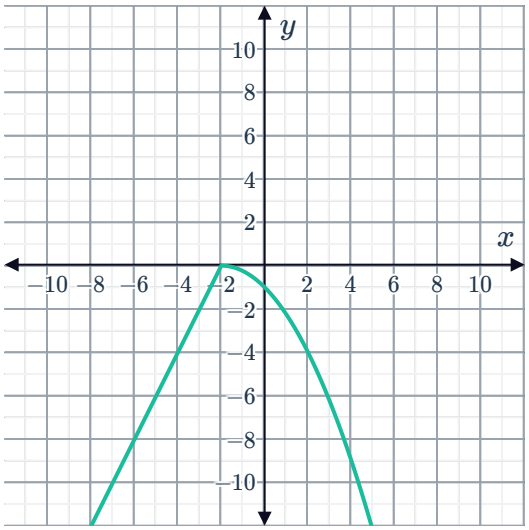

e



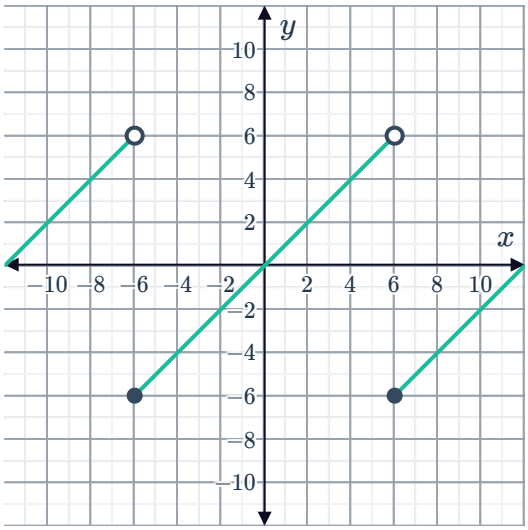

f




g

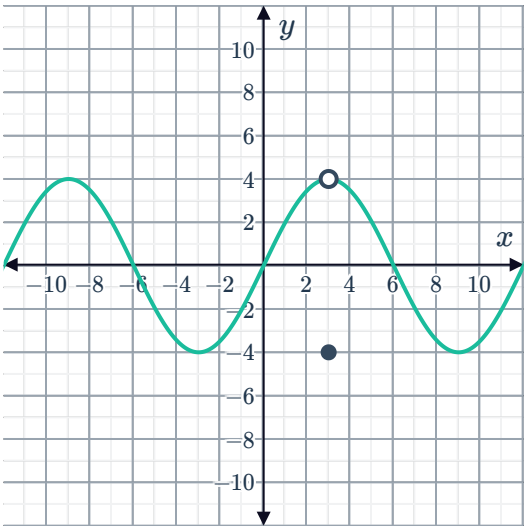


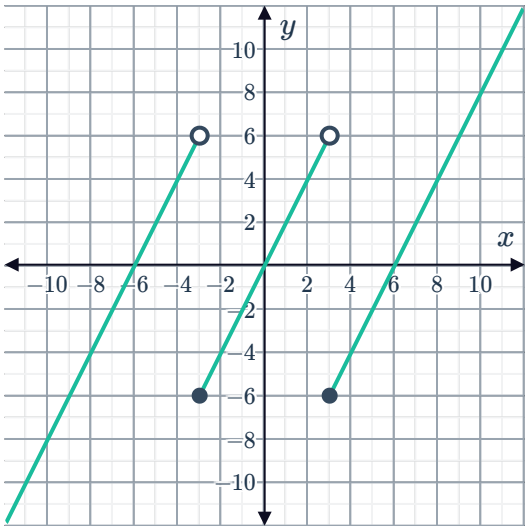

h

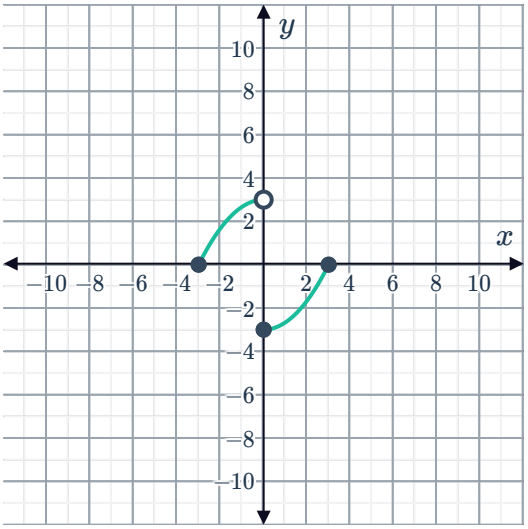



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**14** Consider the following graphs of functions:




State whether the following statements are true or false:

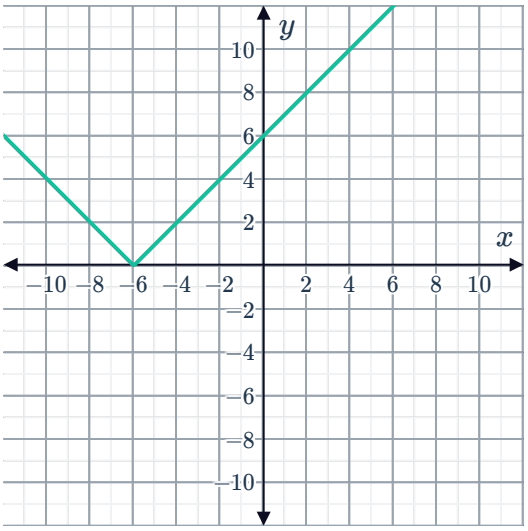
**a** Some of the functions are continuous.

**b** All of the functions have at least one discontinuous point in their domain.

**c** All of the functions have a domain of $(-\infty, \infty)$.

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**15** Consider the graph of the function $f(x) = |x + 6|$:
State whether the function is continuous over the following intervals:

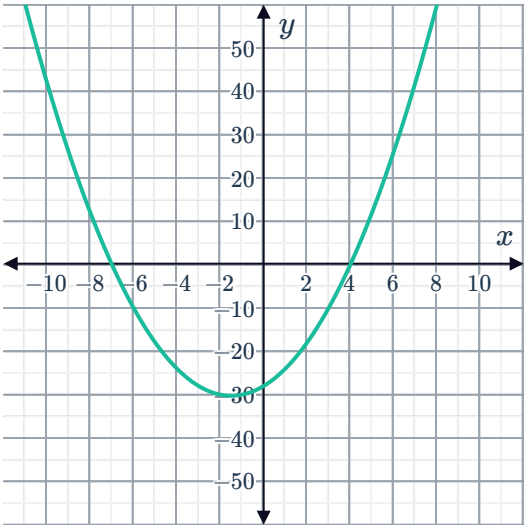



**a** $(-6, \infty)$

**b** $(-\infty, -6)$

**c** $(-\infty, \infty)$

**16** Consider the graph of the function $f(x) = (x + 7)(x - 4)$:
State whether $f(x)$ is a continuous function over:




**a** The domain.

**b** The interval $[-7, 4]$

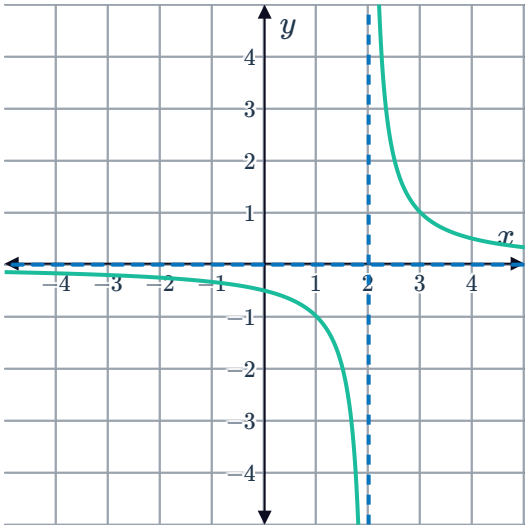
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**17** Consider the graph of the function $f(x) = \dfrac{1}{x-2}$:

**a** State the domain of $f(x)$.

**b** Is $f(x)$ continuous over the interval $(-\infty, 2) \cup (2, \infty)$?

**c** Is $f(x) = \dfrac{1}{x-2}$ a continuous function over its domain?

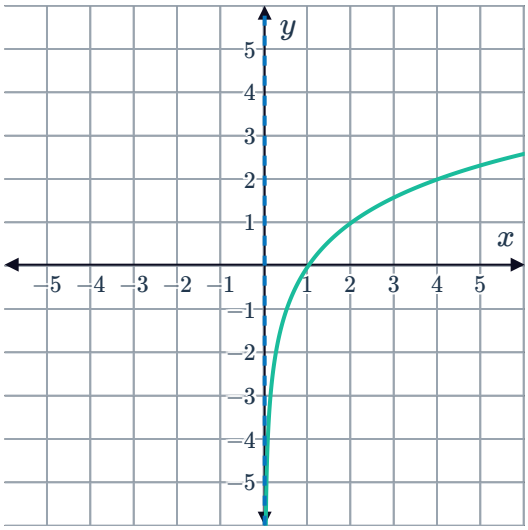



**18** Consider the graph of the function $f(x) = \log_2 x$:

**a** State the domain of $f(x)$.

**b** Is $f(x)$ continuous over the interval $(0, \infty)$?

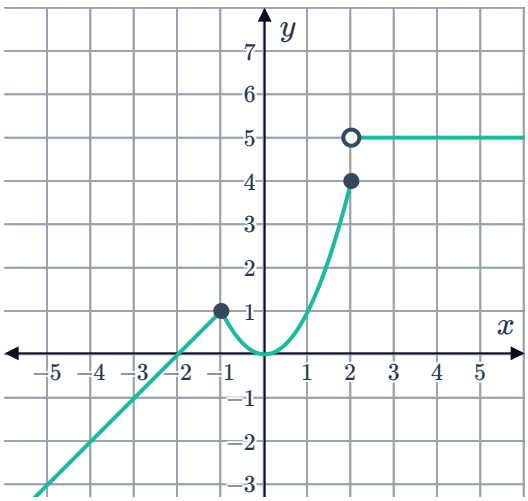
**c** Is $f(x) = \log_2 x$ a continuous function over its domain?




**19** Consider the function $f(x)$ shown:

**a** Is $f(x)$ continuous over its domain $(-\infty, \infty)$?

**b** State the largest interval over which $f(x)$ is continuous.



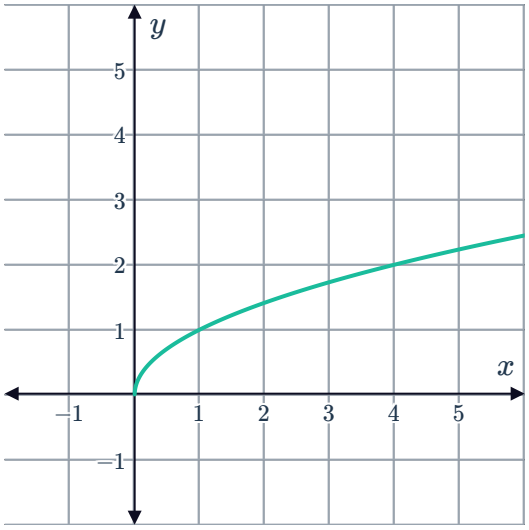

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**20** Consider the graph of the function $f(x) = \sqrt{x}$:

**a** State the domain of $f(x)$.

**b** State the largest interval over which $f(x)$ is continuous.

**c** Is $f(x) = \sqrt{x}$ a continuous function over its domain?

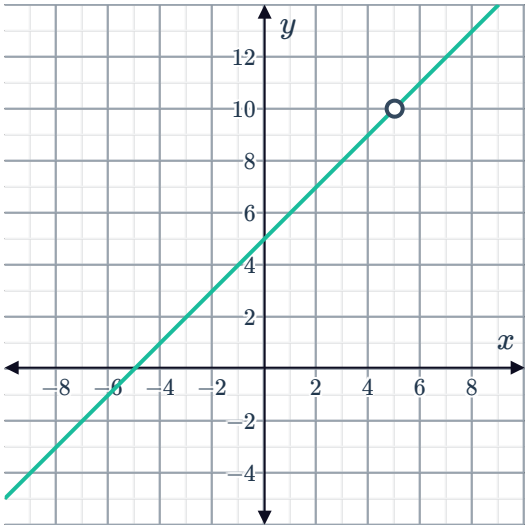



**21** Consider the graph of the function $f(x) = \dfrac{x^2 - 25}{x - 5}$:

**a** State the domain of $f(x)$.

**b** State the largest interval over which $f(x)$ is continuous.

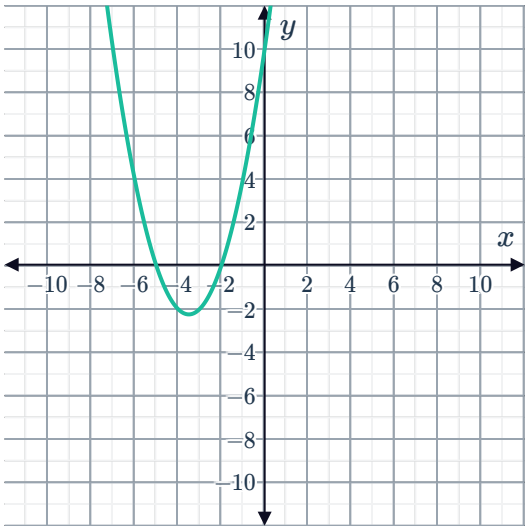
**c** Is $f(x) = \dfrac{x^2 - 25}{x - 5}$ a continuous function over its domain?




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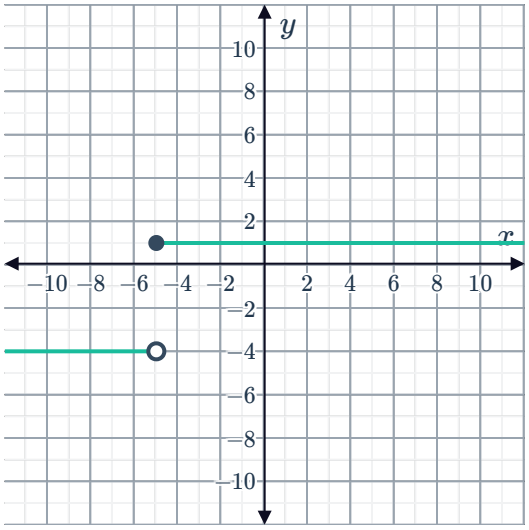
**22** Consider the graph of the function $f(x) = x^2 + 7x + 10$:

- **a** State the domain of $f(x)$.
- **b** State the largest interval over which $f(x)$ is continuous.
- **c** Is $f(x) = x^2 + 7x + 10$ a continuous function over its domain?



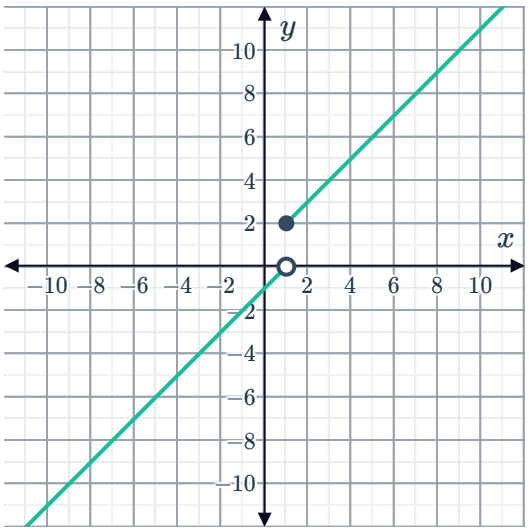

**23** Consider the step function $f(x)$ shown:

- **a** State the domain of $f(x)$.
- **b** State the largest interval over which $f(x)$ is continuous.
- **c** Is $f(x)$ a continuous function over its domain? Explain your answer.




**24** Consider the function $f(x)$ shown:

- **a** State the domain of $f(x)$.
- **b** State the largest interval over which $f(x)$ is continuous.
- **c** Is $f(x)$ a continuous function over its domain? Explain your answer.




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## Applications

**25** The function $f(n)$ is used to determine the area of a square given its side length.

**a** State whether the following values are part of the domain of the function:

**i** $-8$ **ii** $6.5$ **iii** $9\frac{1}{3}$ **iv** $\sqrt{78}$

**b** For $n \geq 0$, state the area function for a side length of $n$.

**c** Sketch the graph of the function $f$.

**26** The number of bees in a colony is initially measured and left to form a hive. The number of bees in the hive is measured each day over the course of one week. The function

$$H(x) = 30(3)^x$$

is found to model the number of bees, $H$, after $x$ days.

**a** What is the domain of the function?

**b** What is the initial number of bees in the colony?

**c** Can the function be used to model the number of bees over an entire season?

**27** The water level in a dam is measured to be $190\ \text{yd}$. After two weeks, the water level is measured again and found to be $140\ \text{yd}$.

**a** Is there a time during the two weeks that the water level of the dam is at $160\ \text{yd}$?

**b** To answer part (a) what property do we assume about the water level over the two weeks?

**c** Consider a function $f(x)$ with domain $[0, 2]$, where $f(0) = 190$ and $f(2) = 140$. If the function is known to be discontinuous, does the function take a value of $160$ at some point in its domain? Explain your answer.

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